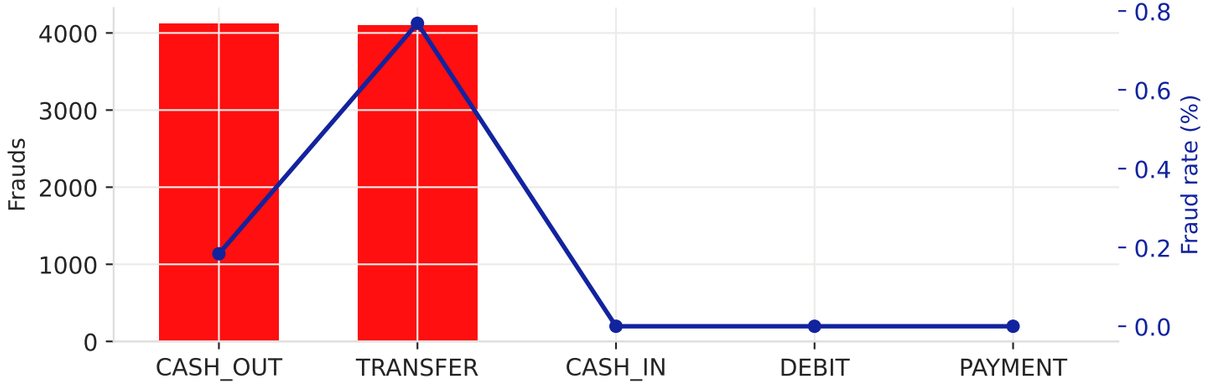
Paysim Fraud Detection Dashboard

6.36 million mobile money transactions | 8,213 frauds | fraud rate 0.129%

Transactions	Frauds	Fraud rate
6,362,620	8,213	0.129%

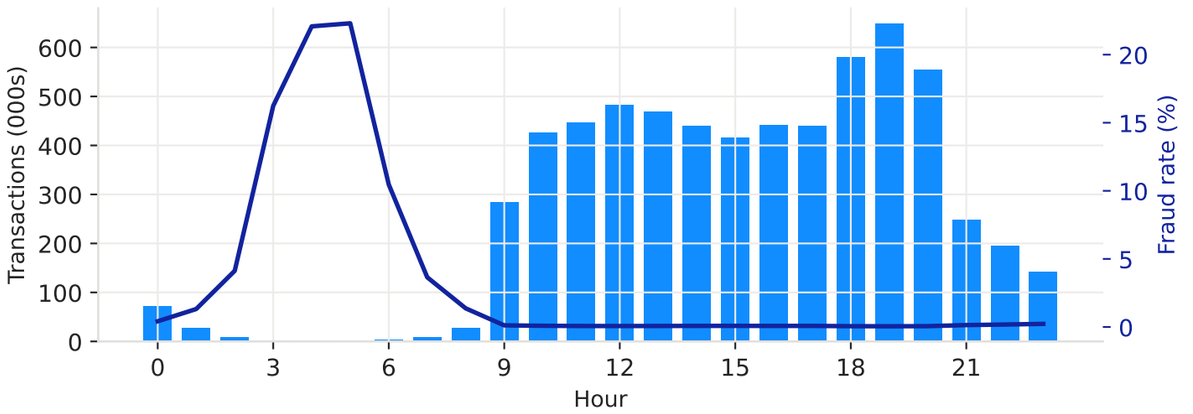
Fraud by transaction type

Columns: fraud count. Line: fraud rate (%). Fraud occurs only in TRANSFER and CASH_OUT.



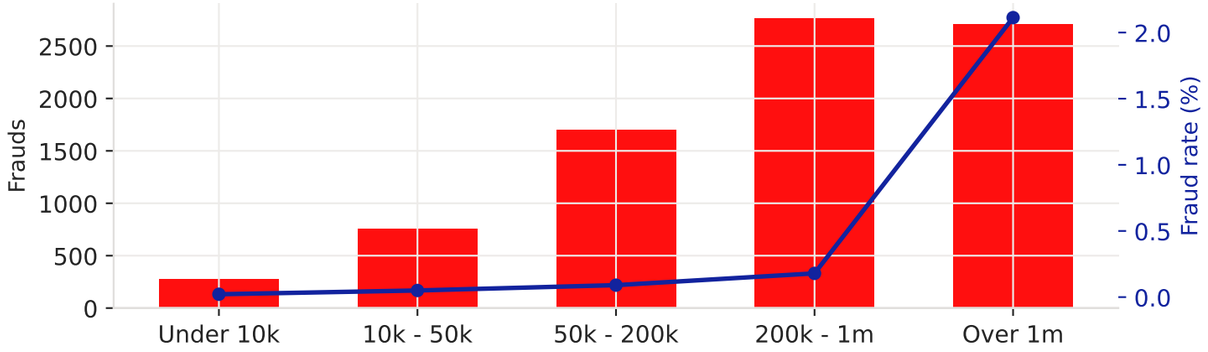
Fraud rate peaks overnight

Fraud volume is flat all day, but legitimate traffic collapses after midnight, so the rate spikes to 22% at 05:00



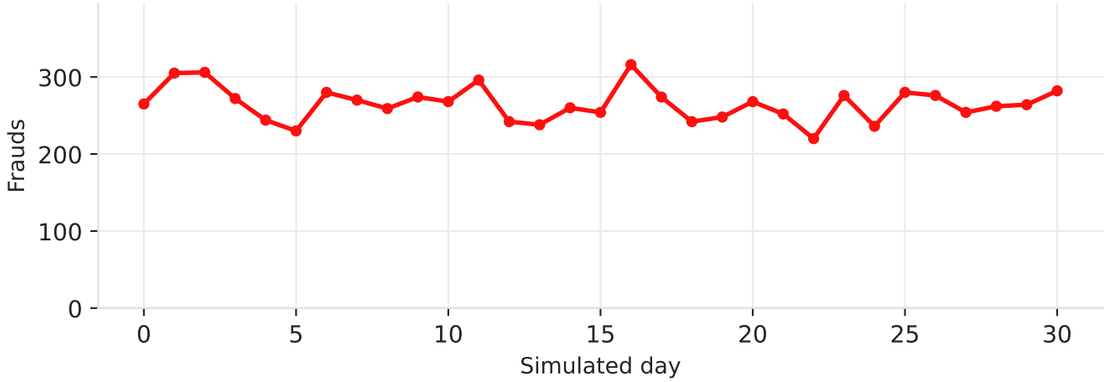
Fraud rate rises sharply with transaction amount

Transactions over 1m are roughly 100 times riskier than those under 10k. Legitimate rows reweighted to full volume.



Fraud count per simulated day

Between 265 and 310 frauds a day across the window, with no trend.

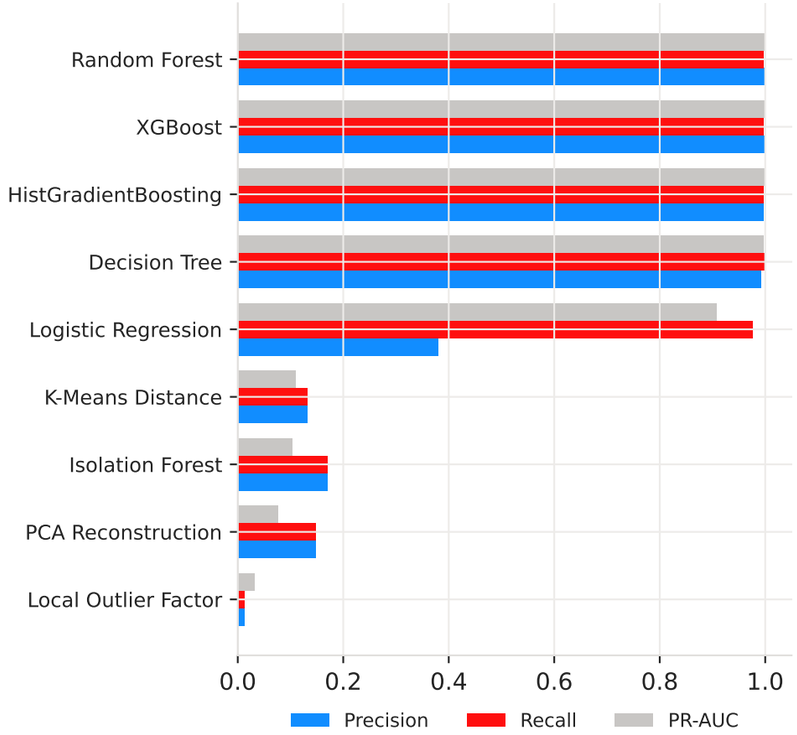


Model Comparison

RQ1 models | RQ2 imbalance strategies | RQ3 anomaly detection

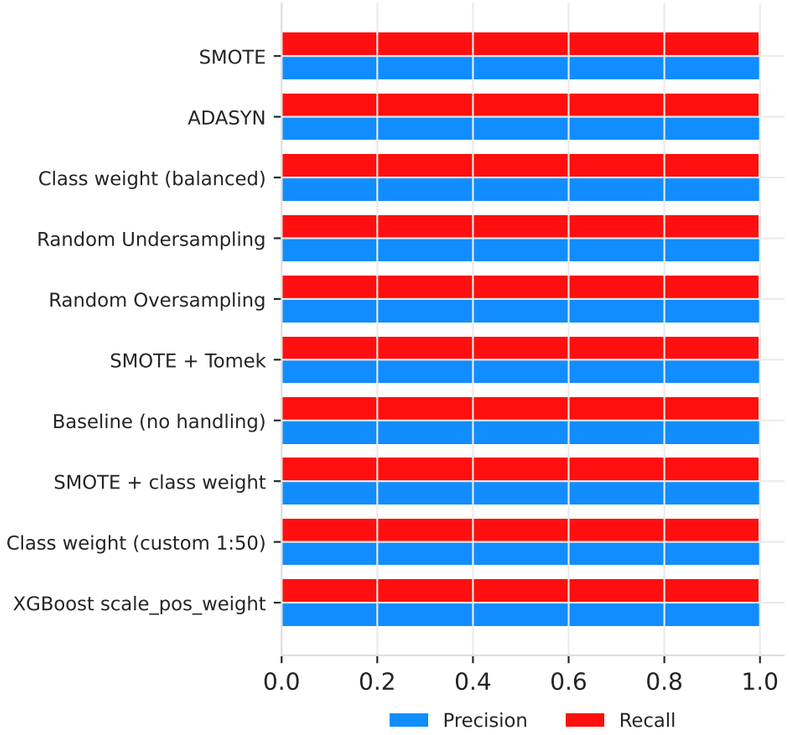
RQ1: supervised versus unsupervised

Accuracy is not shown: always predicting legitimate scores 98%.



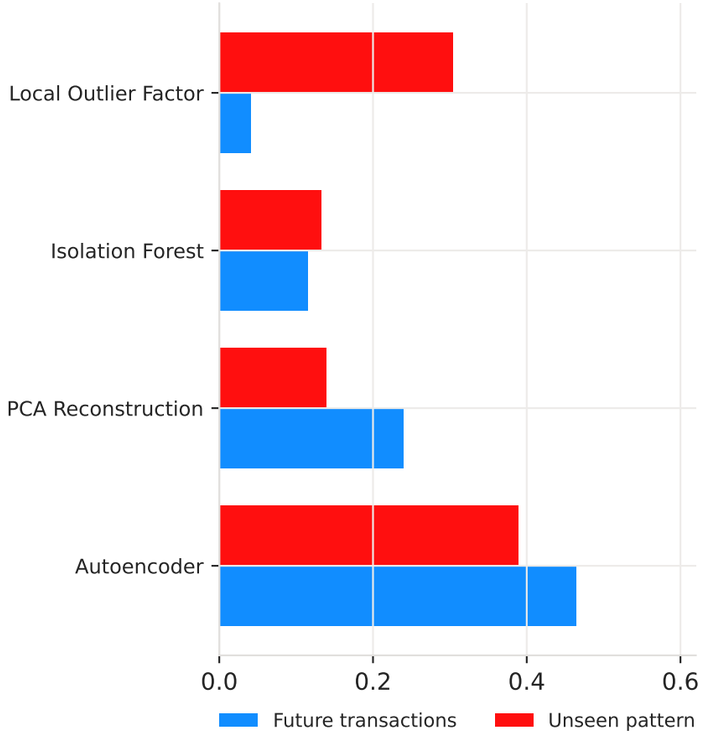
RQ2: imbalance strategies

All ten land within one missed fraud of each other.



RQ3: detectors without labels

PR-AUC on future data and on hidden fraud patterns.



Key findings

RQ1 Supervised wins decisively

Random Forest reaches PR-AUC 0.999 with 8 missed frauds and no false alarms. The best unsupervised method reaches only 0.110.

RQ2 Resampling adds nothing here

All ten strategies land within 0.0008 F1 of each other, a difference of one transaction. Moving the alert threshold to 0.02 cuts expected cost by 77%.

RQ3 Anomaly detection is a complement

The autoencoder is the best label-free detector (PR-AUC 0.46) and degrades least on unseen patterns, but the supervised model still beats it in both scenarios.

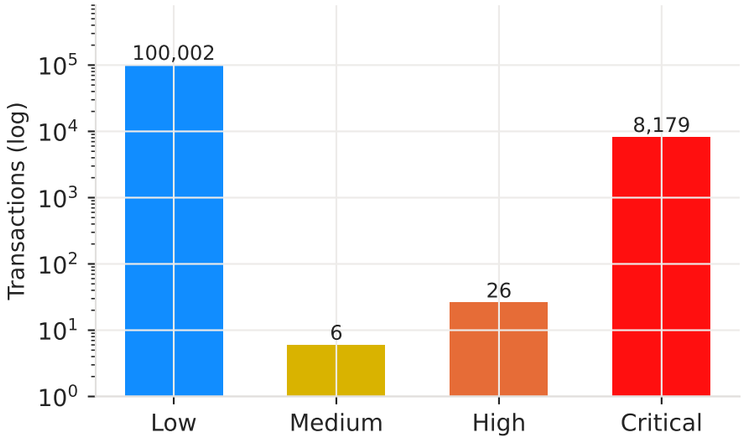
Caveat: PaySim is synthetic, and the balance-error features that drive the model are partly simulator artefacts. Near-perfect scores will not transfer to real fraud.

Flagged Transactions

Scored sample: all 8,213 frauds plus 100,000 legitimate rows | alert threshold 0.02

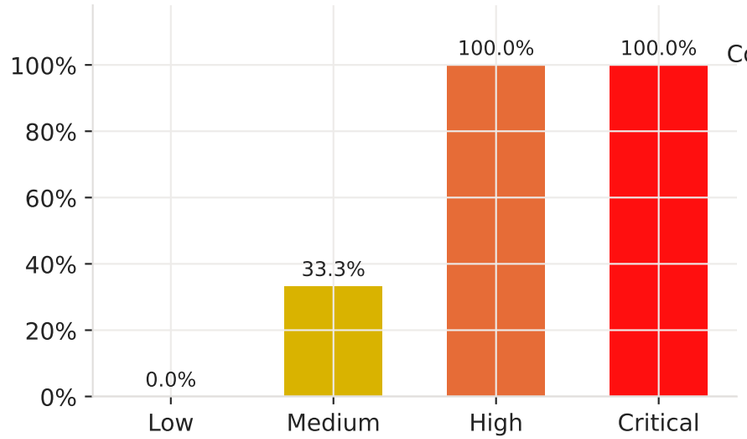
Transactions by risk band

Scored sample, which over-represents fraud by design.



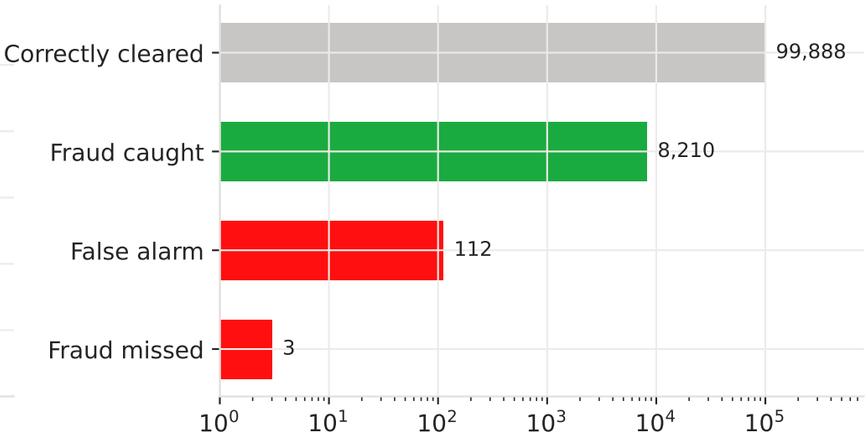
Fraud share inside each band

High and Critical are entirely fraud. Medium holds 6 rows, so its share means little.



Alert outcomes at the chosen threshold

8,210 frauds caught, 3 missed, 112 false alarms.



Top flagged transactions

Txn ID	Type	Amount	Model score	Anomaly score	Band	Actual fraud
TX00001	Cash Out	1,684,039	1.000	0.684	Critical	Yes
TX00002	Cash Out	9,299,230	1.000	0.678	Critical	Yes
TX00003	Cash Out	2,739,248	1.000	0.676	Critical	Yes
TX00004	Cash Out	7,677,121	1.000	0.673	Critical	Yes
TX00005	Cash Out	10,000,000	1.000	0.672	Critical	Yes
TX00006	Cash Out	10,000,000	1.000	0.668	Critical	Yes
TX00007	Cash Out	10,000,000	1.000	0.666	Critical	Yes
TX00008	Cash Out	4,737,833	1.000	0.662	Critical	Yes
TX00009	Cash Out	1,936,404	1.000	0.660	Critical	Yes
TX00010	Cash Out	2,000,718	1.000	0.659	Critical	Yes
TX00011	Cash Out	10,000,000	1.000	0.658	Critical	Yes
TX00012	Cash Out	10,000,000	1.000	0.655	Critical	Yes
TX00013	Cash Out	4,128,854	1.000	0.654	Critical	Yes
TX00014	Cash Out	10,000,000	1.000	0.653	Critical	Yes